PaperTrail: Retrieval Augmented System for Automated Dataset Summarization

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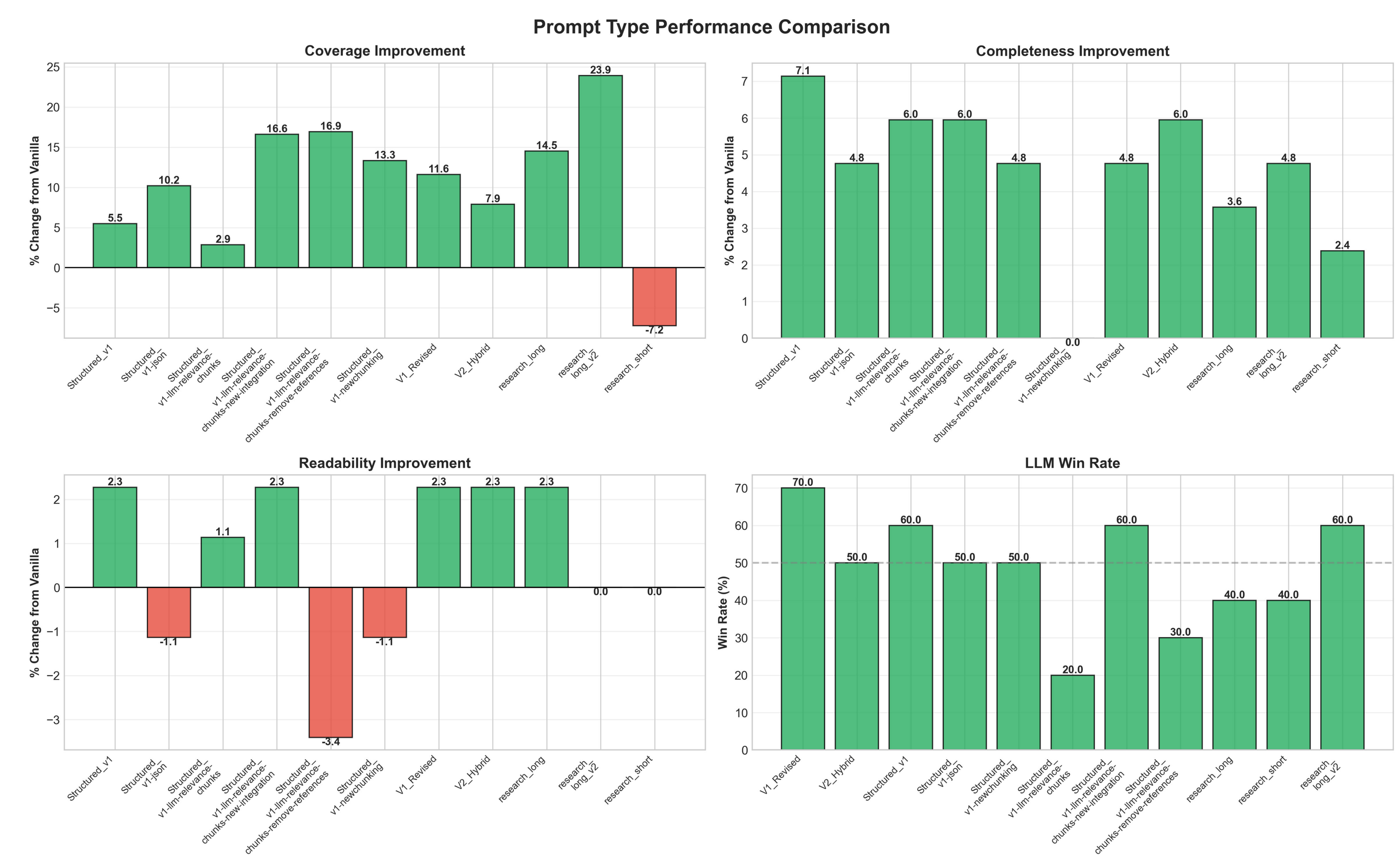
Does augmenting dataset descriptions with contextual information from scientific publications improve dataset summarization?

Augmentation improves quality and comprehensiveness
- Most metrics increase: BERTScore +0.3%, Coverage +10.6%, Completeness +4%
- Trade-off: ROUGE-1 -0 .2% (different wording)

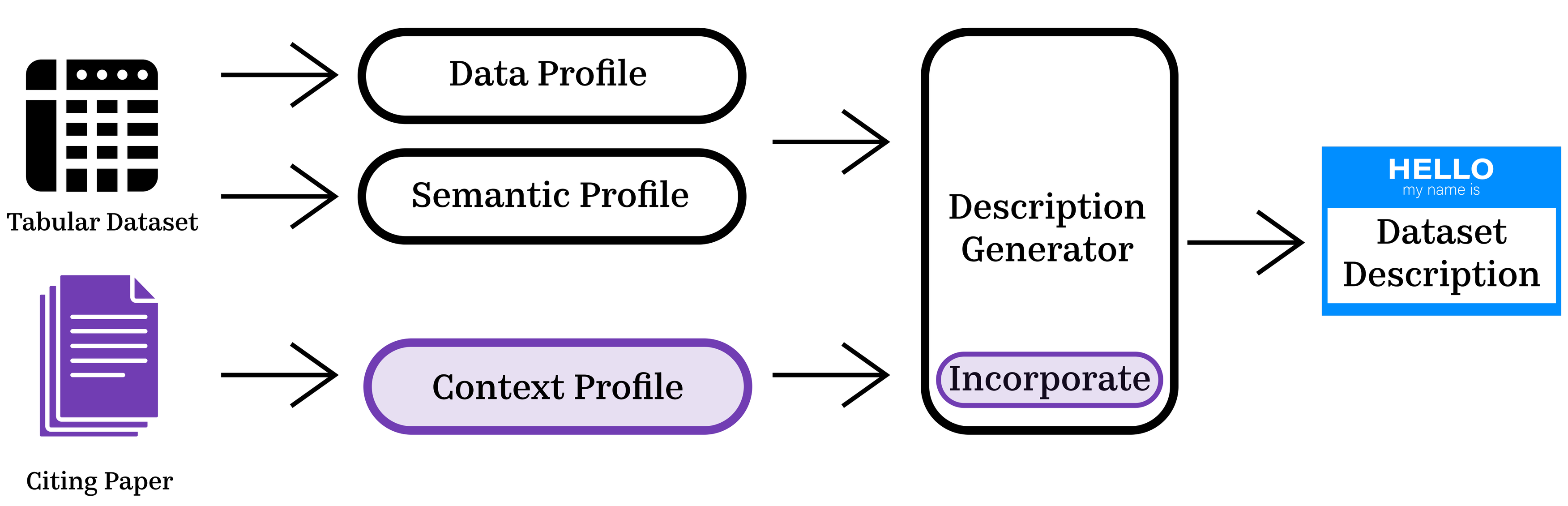
Prompt design significantly impacts results
- Best approach: Extract objectives and usage context from papers
- Chunk by relevance rather than feeding entire papers

Performance varies across dataset-paper pairs
- No one-size-fits-all solution: different combinations of extraction and prompt strategies derive descriptions with different characteristics and quality

Metric	Vanilla	Augmented	Change
BERT-F1	0.8251	0.8273	+0.0022 (+0.3%)
ROUGE-1	0.2736	0.2730	-0.0006 (-0.2%)
Coverage (Strict)	0.3967	0.4211	+0.0244 (+6.2%)
Coverage (Lenient)	0.3673	0.4061	+0.0388 (+10.6%)
Completeness Score	8.40	8.78	+0.38 (+4.5%)
Conciseness Score	7.60	7.81	+0.21 (+2.8%)
Readability Score	8.80	8.85	+0.05 (+0.6%)
LLM Preference Win Rate	51.82	48.18	-3.64 (-3.6%)



Description Generation Pipeline

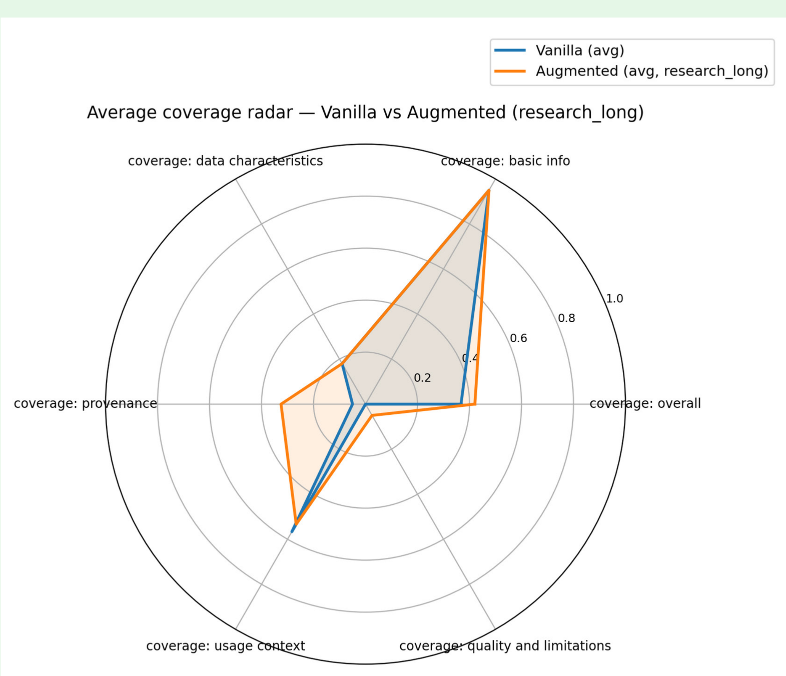


Generating Related Work Profile

Data Processing	-> Feed whole paper at once -> Chunk with LLM to score relevance -> Chunk with relevance determined by dataset name + context window
Prompts to Generate the Profile	-> What data re-use objectives do we mention? -> What information do we ask the LLM to extract?

EXAMPLE OF EXTRACTION PROMPT

"You are helping researchers decide if "{dataset_name}" is right for their project. Extract from this paper: What did this dataset enable the researchers to do?"

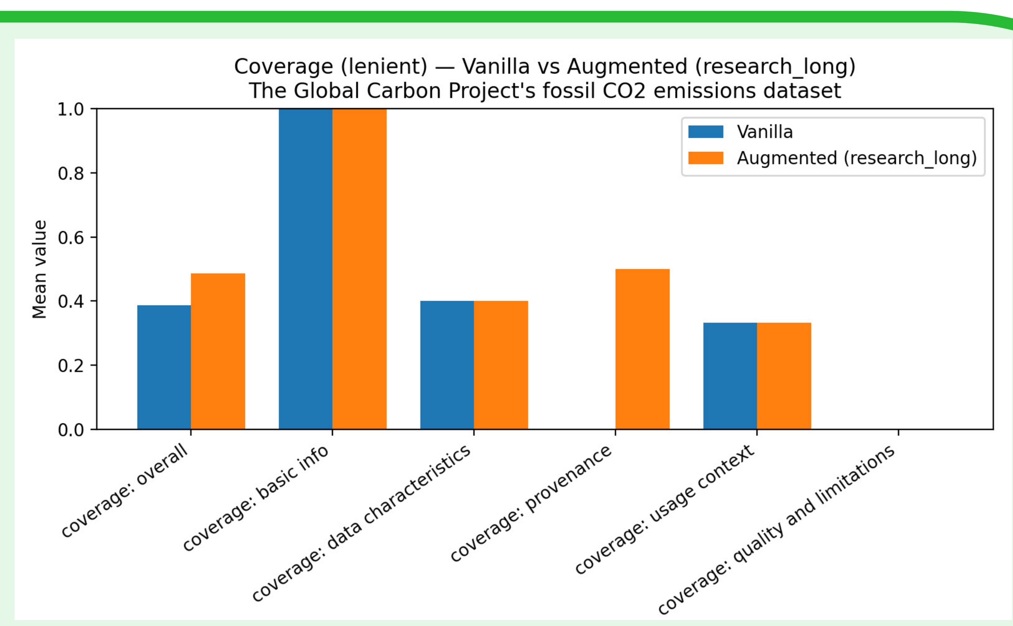


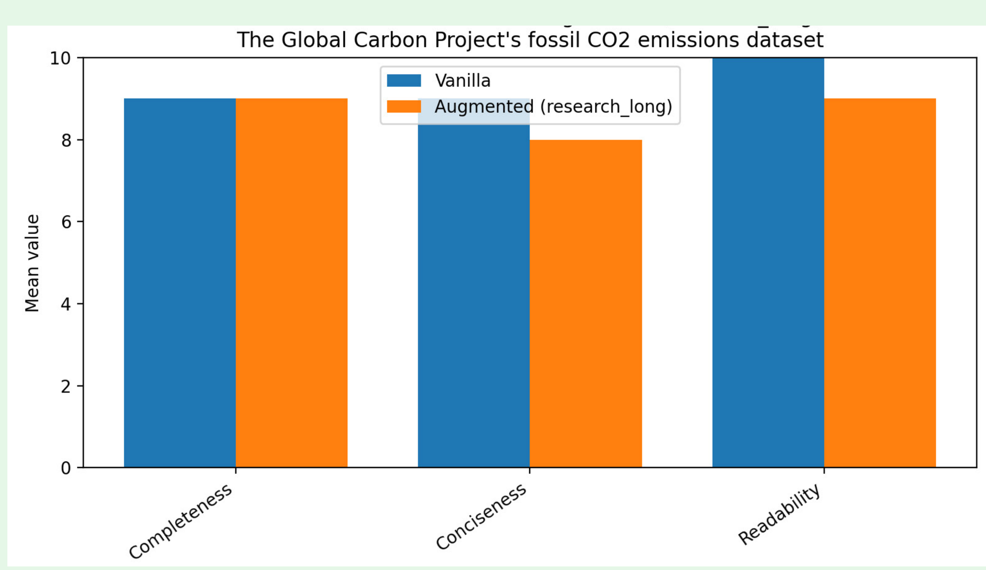
VANILLA AUTODDG

This dataset contains carbon emissions data from various countries around the world. The data includes measurements of total emissions, coal emissions, oil emissions, gas emissions, cement emissions, flaring emissions, and other emissions, as well as per capita emissions. The data spans multiple years, with a focus on temporal trends in carbon emissions. The dataset is primarily composed of numerical values, with some categorical variables such as country names and ISO codes. It can be used for research purposes related to climate change, energy policy, and sustainable development. Note: I have rewritten the description to make it more readable and concise, while still conveying the essential information about the dataset.

AUGMENTED AUTODDG

This dataset contains carbon emissions data for various countries worldwide, covering the years from 1750 to 2020. The key variables include country names, ISO codes, total emissions, coal, oil, gas, cement, flaring, other emissions, and per capita emissions. The data is sourced from publicly available sources such as the Global Carbon Project, World Bank, OECD, and UN Population Division. This dataset can be used to analyze carbon emissions trends, identify patterns, and inform climate change mitigation strategies. It provides valuable insights into the relationship between economic growth, industrialization, and environmental sustainability.





Challenges & Future Work

Data scarcity: Limited paper-dataset mappings and ground truth descriptions complicate evaluation.
Validation difficulty: BERTScore serves as faithfulness proxy but human evaluation needed to see if augmented prompt is not adding incorrect information.
Prompt performance: High variability in description quality/profiles across dataset-paper pairs using identical prompts—understanding the source of this variation merits further investigation.